

SKOVORODINO, Russian Federation

WMO: 306920

Lat: 54.000N

Lon: 123.967E

Elev: 1303

StdP: 14.02

Time Zone: 9.00 (E09)

Period: 94-14

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-40.2	-36.7	-43.8	0.5	-39.7	-40.4	0.6	-36.5	17.3	0.9	14.3	-3.9	3.0	50	0.673

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.9	84.2	67.5	80.9	65.5	77.8	64.4	70.4	79.0	68.6	76.6	66.8	74.4	6.5	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
67.7	106.9	74.2	66.0	100.6	72.4	64.1	94.1	70.5	35.3	78.6	33.8	76.8	32.3	74.4	84.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
17.7	14.3	13.4	-45.5	90.9	3.1	3.0	-47.7	93.0	-49.6	94.8	-51.3	96.4	-53.6	98.6
			-45.5	75.7	3.1	4.3	-47.7	78.8	-49.5	81.3	-51.3	83.7	-53.5	86.8
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	25.8	-15.6	-8.5	7.9	31.2	47.6	60.8	65.2	59.6	46.4	26.7	1.3	-14.5
	DBStd	30.53	9.03	8.02	10.38	9.02	6.89	6.19	5.04	6.17	7.65	10.17	12.36	10.06
	HDD50	10021	2035	1638	1304	567	128	4	0	6	158	721	1460	2000
	HDD65	14403	2500	2058	1769	1015	541	154	59	187	559	1186	1910	2465
	CDD50	1203	0	0	0	1	52	326	472	302	49	1	0	0
	CDD65	113	0	0	0	0	1	27	66	19	0	0	0	0
	CDH74	1620	0	0	0	0	46	536	703	318	17	0	0	0
	CDH80	366	0	0	0	0	6	125	169	66	0	0	0	0
Wind	WSAvg	5.0	4.1	4.6	5.6	6.4	6.8	5.6	4.6	4.6	5.1	5.0	4.3	4.0
Precipitation	PrecAvg	17.1	0.2	0.2	0.3	0.8	1.5	3.0	4.4	3.3	2.3	0.6	0.4	0.2
	PrecMax	21.5	0.5	0.4	0.8	2.1	3.2	7.8	6.9	5.7	6.3	1.9	1.1	0.6
	PrecMin	12.8	0.0	0.0	0.1	0.1	0.1	0.8	1.9	0.7	0.5	0.0	0.1	0.0
	PrecStd	2.6	0.1	0.1	0.2	0.5	0.9	1.7	1.6	1.3	1.5	0.4	0.2	0.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	12.6	22.7	43.1	63.3	79.3	87.7	88.9	85.9	76.4	58.0	31.1	18.4
		MCWB	10.0	18.2	33.8	46.6	57.3	69.3	69.3	69.3	61.5	46.0	26.4	15.2
	2%	DB	8.1	16.7	37.3	56.4	72.9	83.6	84.5	81.3	71.0	51.2	27.2	13.3
		MCWB	6.2	13.9	30.2	42.5	53.8	66.3	68.3	66.2	57.3	41.4	24.8	11.0
	5%	DB	4.4	11.4	31.7	51.3	67.8	80.1	81.2	77.6	66.3	45.8	24.0	7.3
		MCWB	2.9	8.8	26.2	39.2	51.5	63.9	66.7	64.9	55.0	37.7	21.7	5.5
	10%	DB	0.5	7.3	27.6	46.7	63.2	76.6	77.9	73.3	61.5	41.9	20.3	1.2
		MCWB	-0.7	5.3	23.3	36.7	49.1	62.3	66.2	63.3	52.0	35.2	18.2	-0.1
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	10.2	18.3	34.5	47.8	59.7	72.4	73.0	71.6	63.4	46.9	27.8	15.3
		MCDB	12.5	22.0	42.5	61.9	73.7	82.1	81.1	79.8	74.4	56.0	29.5	18.1
	2%	WB	6.2	13.9	30.2	43.3	56.2	68.9	71.0	69.0	59.5	42.5	24.9	11.1
		MCDB	8.0	16.4	36.7	55.0	67.8	79.0	79.4	76.7	67.2	49.5	27.4	13.2
	5%	WB	2.9	9.1	26.6	39.8	53.4	66.3	69.4	66.7	56.5	38.5	21.9	5.4
		MCDB	4.4	11.4	31.2	49.8	64.0	76.9	77.5	74.0	63.1	45.1	24.1	7.2
	10%	WB	-0.6	5.3	23.3	37.1	50.8	63.8	67.9	64.5	53.7	35.3	18.1	-0.1
		MCDB	0.5	7.3	27.6	45.5	61.1	73.6	74.9	71.6	60.7	41.4	20.2	1.1
Mean Daily Temperature Range	5% DB	MDBR	23.6	29.4	30.1	22.9	24.5	25.8	21.9	23.4	24.9	22.0	22.8	20.7
		MCDBR	23.4	34.0	32.9	30.0	34.6	34.0	29.9	31.6	32.8	27.3	22.7	20.1
		MCWBR	21.8	31.1	26.3	17.1	17.8	17.4	15.4	17.5	19.4	17.6	19.4	18.2
	5% WB	MCDBR	23.2	33.9	31.4	27.3	28.8	29.3	22.8	25.4	27.2	25.6	22.1	20.4
		MCWBR	21.8	31.1	25.2	16.1	16.4	16.6	12.8	15.0	17.5	17.0	18.9	18.6
Clear-Sky Solar Irradiance	taub	0.247	0.277	0.326	0.455	0.525	0.482	0.459	0.404	0.357	0.330	0.277	0.249	
	taud	2.142	2.113	2.040	1.862	1.826	2.049	2.171	2.317	2.388	2.286	2.147	2.073	
	Ebn at Noon	226	256	266	241	231	245	249	258	257	234	207	193	
	Edh at Noon	20	30	41	56	62	50	43	36	29	25	20	18	
All-Sky Solar Radiation	RadAvg	377	742	1292	1675	1642	1879	1637	1425	1079	698	416	279	
	RadStd	16	21	50	120	115	168	155	142	89	41	23	14	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (7 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page